

D874

Three Dimensional Science and Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

In a lesson using a free weather app on classroom tablets, learners analyze a year of sunrise times to find a pattern and connect it to Earth’s tilt. Which part of the lesson is the crosscutting concept?

- A. Earth’s tilt causes seasonal changes in daylight
- B. The weather app
- C. Analyzing and interpreting data
- D. Patterns

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

“Developing and using models” belongs to which dimension?

- A. A teaching strategy the teacher uses
- B. Crosscutting concepts
- C. Science and engineering practices
- D. Disciplinary core ideas

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teacher candidate is choosing an anchor for a chemical-reactions unit that runs for three weeks in October. Which choice best meets the qualities of an anchoring phenomenon?

- A. The textbook definition of a chemical reaction
- B. Ice cubes melting in a glass of water
- C. The largest explosion video available, since more dramatic is always better
- D. A silver spoon in a kitchen drawer that slowly turns black over several weeks

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A candidate brainstorms phenomena for an ecosystems unit. Her mentor's room has a large aquarium and a reading corner with beanbags. Which item on her list is a phenomenon?

- A. A lake turned bright green with algae in late summer.
- B. Ecosystems
- C. Energy flows through food webs.
- D. The classroom aquarium

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A lesson plan lists “photosynthesis” as its phenomenon. What is the best critique?

- A. It is fine, because anything that happens in nature is a phenomenon.
- B. It names the concept to be learned, not an event to explain, such as a plant gaining mass while the soil barely changes.
- C. It is not a phenomenon because phenomena must be surprising explosions or color changes.
- D. It becomes a phenomenon once learners do a lab on it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Using new digital temperature probes, learners follow a worksheet to record how fast a hot drink cools in two kinds of cups. The lesson ends when the data table is full. How should the candidate make it three-dimensional?

- A. Add more measurement steps and a third cup.
- B. Nothing; hands-on activities always include all three dimensions.
- C. Have learners use their data to explain, as cause and effect, how energy moves from the drink through each cup.
- D. Add a vocabulary quiz on conduction and insulation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A lesson has learners explain why their drink cooled faster in one cup using energy transfer, but they never collect or analyze any data or build a model. Which dimension is missing?

- A.** None; explaining is enough for three-dimensional learning.
- B.** The disciplinary core idea
- C.** The crosscutting concept
- D.** A science and engineering practice

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

After the anchor, learners post their questions on a bulletin board decorated for fall. The candidate's mentor says the board can take up a lot of wall space. How should the candidate use the driving question board?

- A.** Take the board down to save wall space once questions are posted.
- B.** Answer every question on the first day so no one is confused.
- C.** Group the questions, choose which to investigate next, and revisit the board to track what has been figured out.
- D.** Replace learners' questions with the teacher's own list.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Learners will design an insulated lunch box. A local hardware store donated foam and foil, and teams will present in the gym. What should teams do first?

- A.** Build one prototype, test it once, and call it final.
- B.** Define the problem: criteria (keep food below 5 °C for 4 hours) and constraints (materials, size, time).
- C.** Start building right away with the donated materials.
- D.** Pick the thickest foam, because the donated materials decide the design.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A team's lunch-box prototype keeps food cold for only 2 of the required 4 hours. What should the team do next?

- A.** Change the criterion to 2 hours to match the result.
- B.** Analyze where heat got in, revise the design, and retest against the 4-hour criterion.
- C.** Accept it, since engineering designs are never successful on the first try anyway.
- D.** Abandon the design and try something random.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A candidate was observed by a university supervisor on a rainy Friday. Which reflection note shows the strongest evidence-based practice?

- A.** "About half the learners said metal feels cold because it holds 'coldness'; next lesson I will have them measure both blocks with probes."
- B.** "Learners always struggle with heat, so there is nothing to change."
- C.** "My slides looked professional and my timing was good."
- D.** "The lesson went well and learners were engaged."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A candidate found a \$300 kit for building model wind turbines in the department storeroom. She is excited to build her energy unit around it. Where should her planning begin?

- A.** With the performance expectation and the evidence that will show learners met it, then deciding whether the kit fits.
- B.** With the kit, then finding standards that match it.
- C.** With whatever else is in the storeroom, since available materials decide the unit.
- D.** With writing the unit test after the unit is taught.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

In a lesson ~~using a free weather app on classroom tablets~~, learners analyze a year of sunrise times to find a pattern and connect it to Earth's tilt. Which part of the lesson is the crosscutting concept?

Target word: identify the crosscutting concept **Decides:** Patterns is a crosscutting concept.

A	Adjacent category	That is a disciplinary core idea.
B	Camouflage	A tool, not a dimension.
C	Confusion set	That is a science and engineering practice.
D	Correct	Patterns is one of the seven crosscutting concepts.

2. Answer C Concept 1 • transfer

"Developing and using models" belongs to which dimension?

Target word: classify a named element **Decides:** It is something learners do.

A	Inverted operation	Reverses who does it; learners develop the models.
B	Confusion set	"Systems and system models" is a crosscutting concept, but developing models is a practice.
C	Correct	Modeling is one of the eight practices.
D	Adjacent category	Core ideas are content, not actions.

3. Answer D Concept 2

A teacher candidate is choosing an anchor for a chemical-reactions unit ~~that runs for three weeks in October~~. Which choice best meets the qualities of an anchoring phenomenon?

Target word: best anchoring phenomenon **Decides:** Observable, puzzling, familiar, and rich enough for a unit.

A	Adjacent category	A definition is a topic, not an event to explain.
B	Dropped element	Learners can already explain it, and it is a physical change.
C	Absolute language	Drama does not ensure fit or depth.
D	Correct	Familiar, puzzling, and explained by core ideas the unit will build.

4. Answer A Concept 3

A candidate brainstorms phenomena for an ecosystems unit. ~~Her mentor's room has a large aquarium and a reading corner with beanbags~~. Which item on her list is a phenomenon?

Target word: identify a phenomenon **Decides:** A phenomenon is a specific observable event.

A	Correct	An observable event learners can work to explain.
B	Confusion set	A topic label, not an event.
C	Adjacent category	A core idea learners should reach, not the event that drives inquiry.
D	Camouflage	An object in the room, not an event to explain.

5. Answer B Concept 3 · transfer

A lesson plan lists “photosynthesis” as its phenomenon. What is the best critique?

Target word: why a topic is not a phenomenon **Decides:** Photosynthesis is the explanation, not an observable event.

A	Absolute language	Naming a process does not give learners something to puzzle over.
B	Correct	Replace the concept with an observable event it explains.
C	Confusion set	Everyday events can be strong phenomena.
D	Adjacent category	A lab is an activity; the anchor still needs to be an event.

6. Answer C Concept 4

~~Using new digital temperature probes,~~ learners follow a worksheet to record how fast a hot drink cools in two kinds of cups. The lesson ends when the data table is full. How should the candidate make it three-dimensional?

Target word: adding the missing dimensions **Decides:** Learners must use the data and a lens to explain a core idea.

A	Dropped element	More data collection still leaves out sensemaking.
B	Absolute language	Hands-on does not guarantee three-dimensional learning.
C	Correct	Adds the crosscutting concept and core idea to the practice.
D	Adjacent category	Recall does not integrate practices or crosscutting concepts.

7. Answer D Concept 4 · transfer

A lesson has learners explain why their drink cooled faster in one cup using energy transfer, but they never collect or analyze any data or build a model. Which dimension is missing?

Target word: name the missing dimension **Decides:** No practice is used.

A	Absolute language	All three dimensions must work together.
B	Inverted operation	Energy transfer, the core idea, is present.
C	Confusion set	Cause and effect / energy flow is present in the explanation.
D	Correct	Learners reason about content without doing a practice.

8. Answer C Concept 5

After the anchor, learners post their questions on a bulletin board ~~decorated for fall. The candidate's mentor says the board can take up a lot of wall space.~~ How should the candidate use the driving question board?

Target word: purpose of the driving question board **Decides:** Learner questions organize and motivate the sequence.

A	Camouflage	The space issue does not decide the board's purpose.
B	Inverted operation	Gives answers before learners investigate.
C	Correct	Keeps learner questions driving the unit.
D	Dropped element	Removes learner ownership of the inquiry.

9. Answer B Concept 6

Learners will design an insulated lunch box. ~~A local hardware store donated foam and foil, and teams will present in the gym.~~ What should teams do first?

Target word: first step of engineering design **Decides:** Define the problem with criteria and constraints.

A	Absolute language	Design depends on iteration.
B	Correct	Criteria and constraints guide every later design decision.
C	Dropped element	Skips defining what success means.
D	Camouflage	Materials available are a constraint, not the first step.

10. Answer B Concept 6 • transfer

A team's lunch-box prototype keeps food cold for only 2 of the required 4 hours. What should the team do next?

Target word: response to a failed criterion **Decides:** Iterate: analyze, revise, retest.

A	Inverted operation	Moves the goal to fit the design instead of the reverse.
B	Correct	Iteration improves the design against fixed criteria.
C	Absolute language	Imperfection is expected, but the design still must meet the criterion.
D	Adjacent category	Ignores test data that shows what to fix.

11. Answer A Concept 7

A candidate was observed ~~by a university supervisor on a rainy Friday.~~ Which reflection note shows the strongest evidence-based practice?

Target word: best reflection **Decides:** It cites learner evidence and a next step.

A	Correct	Specific learner evidence leads to a planned instructional change.
B	Absolute language	Dismisses evidence instead of using it.
C	Adjacent category	Focuses on the teacher, not learning.
D	Dropped element	No evidence of learner thinking.

12. Answer A Concept 8

A candidate found a ~~\$300~~ kit for building model wind turbines ~~in the department storeroom.~~ She is excited to build her energy unit around it. Where should her planning begin?

Target word: starting point for planning **Decides:** Backward design starts from the performance expectation.

A	Correct	Goals and evidence come before activities.
B	Inverted operation	Reverses backward design.
C	Camouflage	Materials support goals; they do not set them.
D	Adjacent category	Evidence of learning should be planned before instruction.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Naming the three dimensions	Science and engineering practices are what learners do (model, analyze data, argue); crosscutting concepts are lenses across fields (patterns, cause and effect, energy and matter); disciplinary core ideas are the content.	Calling a practice such as analyzing data a crosscutting concept.	“Which part of this lesson is the crosscutting concept?”
2	Qualities of an anchoring phenomenon	A good anchor is observable, puzzling to learners, connected to their experience, and complex enough to drive a whole unit.	Choosing something learners can already explain, or the most dramatic video regardless of fit.	Candidate anchors listed for a unit.
3	Phenomenon vs. topic	A phenomenon is a specific observable event to explain; a topic or concept name is not.	Listing a topic such as “ecosystems” or a process name as the phenomenon.	“Which of these is a phenomenon?”
4	Three-dimensional learning	Learners use a practice and a crosscutting concept to make sense of a core idea; doing a hands-on task alone is not three-dimensional.	Assuming any hands-on activity integrates all three dimensions.	An activity where learners measure or follow steps but never explain.
5	Learner questions drive the sequence	A driving question board collects learner questions about the anchor and is revisited to motivate and track each next investigation.	Answering all learner questions at once or replacing them with teacher questions.	Learners generate questions after the anchor.
6	Engineering design: define, test, iterate	Design begins by defining the problem with criteria and constraints, then cycles through testing and revising against those criteria.	Building first, or testing once and calling it done.	A design challenge with a target and limits.
7	Evidence-based reflection	Useful reflection cites evidence of learner thinking and names a specific next step.	Reflecting on engagement, behavior, or the teacher’s performance without learner evidence.	Candidate reflection notes compared.
8	Plan backward from the performance expectation	Start with the performance expectation, decide what evidence will show it, then choose learning experiences.	Building a unit around a favorite activity first.	“Where should planning begin?”

Course scope (public description): Three Dimensional Science and Engineering focuses on developing a comprehensive understanding of science and engineering pedagogical knowledge. This course is the first of three science teaching methods courses and provides a robust foundation in integrating disciplinary core ideas, crosscutting concepts, and science and engineering practices in phenomena-based curriculum and instruction. Candidates will delve into planning learning experiences, designing instructional strategies, and utilizing phenomena-based teaching to promote engagement and understanding. Reflective practices, such as evaluating observed teaching, analyzing personal teaching methods, and reviewing course content, are emphasized to foster continuous improvement. The course will be assessed via an integrated performance assessment task, requiring

candidates to demonstrate their instructional skills.